CHAPTER 03

Alligations

The concept of Alligations is simply an extension of Averages. In fact, Allegation method is used to find the weighted average of mixture or different groups. It is also used to find out the percentage of impurity or reduction in the original quantity where the repeated dilution or depreciation takes place.

Exams, such as CAT, XAT, IIFT, CMAT and GMAT, ask the application based questions from this chapter. Though the number of problems asked in these exams is not very high, but it helps you fetch good marks in the exam if you understand the Alligation method really well. Nonetheless, exams such SSC CGL and Bank PO can ask good number of problems from this chapter. As far as its scope is concerned, we can use it in the chapters such as Simple and Compound interest, Profit and Loss, Ratio and Proportion to name just a few. In fact, it plays a very significant role in answering the problems based on Data Interpretation.

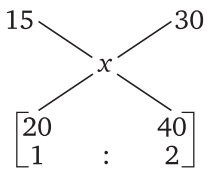
Exp. 1) The average weight of a class of 40 students is 30 and the average weight of a class of 20 students is 15. Find the average weight of both the combined classes :

(a) 20 (b) 25 (c) 17.5 (d) 15

Solution To find the solution of this problem, we can use two widely used techniques :

1. Weighted average method The required average $= \frac{40 \times 30 + 20 \times 15}{(40 + 20)} = 25$

2. Alligation method



In this graphical representation of the solution, x is the weighted average where 15 and 30 are the averages of different class. So remember that the weighted average is always written in middle and the individual averages are written at the top, the smaller one in the left and the greater one in the right just for convenience and the number of elements (or the fixed quantities) are written below correspondingly. Now, the difference between the two averages is divided in the inverse ratio of the quantities written below. As in the above problem, the difference $15\ (=30-15)$ will be divided in the ratio of $2:1$ (as $40:20$) but not in the ratio of $1:2$, which must be clear from the arrows indicating cross proportion. So, the value of $x = 15 + \frac{2}{3} \times (30-15) = 25$ or $x = 30 - \frac{1}{3} \times (30-15) = 25.$

Therefore, the average weight of both the classes is 25.

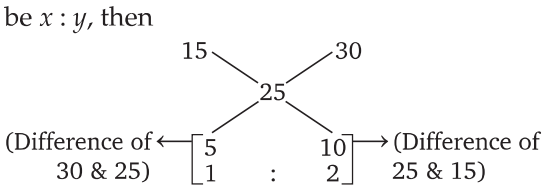
Chapter Checklist

- *Problem Based Mixtures and Groups*
- *Problems Based on Time, Speed and Distance*
- *Problems Based on Dilution and Depreciation*
- ***CAT Test***

Exp. 2) If the average weight of a class of students is 15 and the average weight of another class of students is 30, then find the ratio of the students of the first class to the another class of 30 students when the average weight of both the classes is 25 :

(a) 1 : 2 (b) 2 : 1 (c) 1 : 3 (d) 3 : 4

Solution Let the ratio of the students with 15 students to 30 students be $x : y$, then



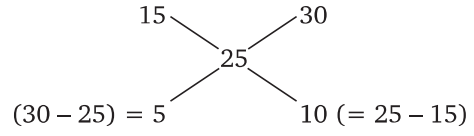
Here we know that the averages of individual classes is 15 and 30. Again we know that the averages of both the combined classes is 25. So the difference between 25 and 15 *i.e.*, 10 and 30 & 25, *i.e.*, 5 is written diagonally opposite.

Thus the required ratio $= \frac{5}{10} = \frac{1}{2}$

Exp. 3) The average weight of girls is 15 and the average weight of boys is 30 and the average weight of boys and girls both is 25. If the number of boys are 12, then the number of girls is :

(a) 4 (b) 6 (c) 10 (d) 18

Solution



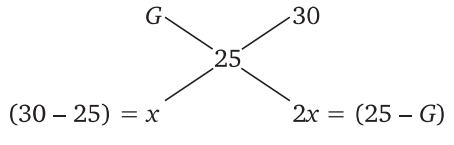
Since the ratio of girls to boys is $\frac{5}{10} = \frac{1}{2}$

Hence if there are 12 boys, so there will be 6 girls.

Exp. 4) The ratio of number of girls to number of boys is 1 : 2. If the average weight of the boys is 30 kg and the average weight of both the boys and girls be 25 kg, then the average weight of the girls is :

(a) 15 kg (b) 20 kg (c) 35 kg (d) 40 kg

Solution



$\Rightarrow \quad \left(\frac{30-25}{25-G} = \frac{x}{2x}\right)$

So, $\frac{(30-25)}{(25-G)} = \frac{1}{2} \Rightarrow G = 15$

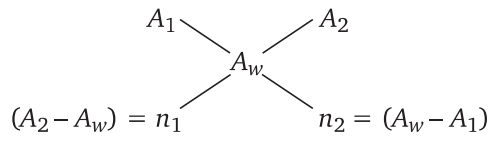
Therefore, the average weight of the girls is 15 kg. In general, if the average of group 1 be A_1 and the number of the elements be n_1 and the average of the group 2 be A_2 and the number of element be n_2 then the weighted average $A_w = \frac{n_1A_1 + n_2A_2}{n_1 + n_2}$

$\Rightarrow \quad n_1A_w + n_2A_w = n_1A_1 + n_2A_2$

$\Rightarrow \quad n_1(A_w - A_1) = n_2(A_2 - A_w)$

$\Rightarrow \quad \frac{n_1}{n_2} = \frac{(A_2 - A_w)}{(A_w - A_1)}$ (known as Alligation Equation)

So, this can be represented in the graphical form as given below

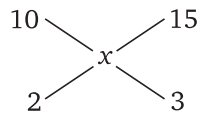


So, all these problems concerned to the topic can be solved either by the formula of weighted average or Alligation equation or graphical representation method, as discussed above.

Exp. 5) Two varieties of soda water with different prices is mixed in the ratio of 2 : 3. The price of first soda water is ₹ 10 per litre while the price of second soda water is ₹ 15 per litre, respectively. The average price of the mixture (per litre) is :

(a) ₹ 12 (b) ₹ 13 (c) ₹ 14 (d) ₹ 15

Solution



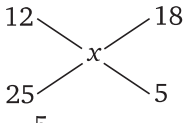
$\Rightarrow \quad \frac{(x-10)}{(15-x)} = \frac{3}{2} \Rightarrow 2(x-10) = 3(15-x)$

$\Rightarrow \quad x = 13$

Exp. 6) 5 kg of superior quality of sugar is mixed with 25 kg of inferior quality sugar. The price of superior quality and inferior quality sugar is ₹ 18 and ₹ 12 respectively. The average price per kg of the mixture is :

(a) ₹ 13 (b) ₹ 15 (c) ₹ 18 (d) ₹ 21

Solution



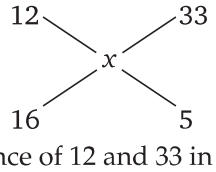
$\frac{(x-12)}{(18-x)} = \frac{5}{25} \Rightarrow x = 13$

NOTE **SHORTCUT** The corresponding ratio is 25 : 5 = 5 : 1
Now reverse the ratio, which is 1 : 5
Now divide the differences of 12 and 18 in the ratio of 1 : 5
i.e., divide 6 into two parts in the ratio of 1 : 5.
Then the average price of mixture is 12 + 1 = 13 or 18 – 5 = 13

Exp. 7) 16 litres of kerosene is mixed with 5 litres of petrol. The price of kerosene is ₹ 12 per litre and the price of petrol is ₹ 33 per litre. The average price of the mixture per litre is :

(a) ₹ 15 (b) ₹ 17 (c) ₹ 23 (d) ₹ 27

Solution



Now divide the difference of 12 and 33 in the ratio of 5 : 16 (not in the ratio of 16 : 5)

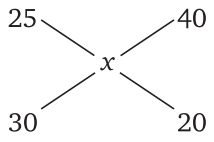
Here, the difference of 33 and 12 = 21
Again on dividing 21 in the ratio of 5 : 16, we get the first part = 5 and second part = 16
So, the average price = 12 + 5 = 17 or = 33 − 16 = 17

NOTE This cross among the numbers written above shows that the difference between *x* and 12 is 5 and the difference between 33 and *x* is 16.

Exp. 8) Amit travels 30 minutes at the speed of 25 km/hr. Further he travels 20 minutes at the speed of 40 km/hr. Find his average speed.

(a) 25 km/hr (b) 30 km/hr
(c) 31 km/hr (d) none of these

Solution



Since the actual ratio is 3 : 2, so reverse it, *i.e.,* the required ratio is 2 : 3. Now divide the difference of both the speeds *i.e.,* 25 and 40 in the ratio of 2 : 3.

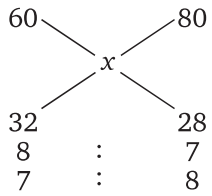
i.e., $(40-25)\times\frac{2}{5}=6$ and $(40-25)\times\frac{3}{5}=9$

So, the average speed is 25 + 6 = 31 or 40 − 9 = 31

Exp. 9) A milkman has two types of milk. In the first container the percentage of milk is 80% and in the second container the percentage of milk is 60%. If he mixes 28 litres of milk of the first container to the 32 litres of milk of the second container, then the percentage of milk in the mixture is :

(a) 63.99 (b) 69.33 (c) 72.5 (d) 75.2

Solution



8 : 7
7 : 8 (reversed ratio)

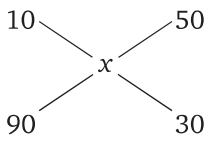
So, $(80-60)\times\frac{7}{7+8}=20\times\frac{7}{15}=9.33$

Thus, the required percentage = 60 + 9.33 = 69.33

Exp. 10) Modern electronic shop sold the 30% hardware at the profit of 50% and 90% software at the profit of 10%. The average profit per cent of the Modern electronic shop is, if it sells only these two kinds of things :

(a) 15 (b) 20 (c) 25 (d) 45

Solution



The actual ratio is 3 : 1. So, the reversed ratio is 1 : 3.
Now divide the difference of 10 and 50 in the ratio of 1 : 3 then add the first part to the 10 or subtract the second part from the 50.

i.e., Required average $=10+(50-10)\times\frac{1}{4}$

$=10+10=20$

or $50-(50-10)\times\frac{3}{4}=20$

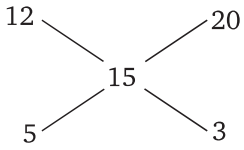
NOTE All the problems discussed above can be solved in any way, *i.e.,* either by using the formula of weighted average or Alligation Equation. But we have emphasised on the graphical (cross proportion) method. Since this technique is quite handy, *i.e.,* some times you need not to calculate on paper.

Exp. 11) Avinash covered 150 km distance in 10 hours. The first part of his journey he covered by car, then he hired a rickshaw. The speed of car and rickshaw is 20 km/hr and 12 km/hr respectively. The ratio of distances covered by car and the rickshaw respectively are :

(a) 2 : 3 (b) 4 : 5
(c) 1 : 1 (d) none of these

Solution The average speed of Avinash

$=\frac{150}{10}=15$ km/hr



It means the rickshaw took 5/8 and car took 3/8 of the total time *i.e.,* the ratio of time taken by rickshaw to car is 5 : 3.
So the ratio of distances covered by rickshaw to car is

$5\times12:3\times20 \Rightarrow 1:1$

NOTE

1. In this calculation (*i.e.,* in alligation method) distance never involves directly. Only time and speeds are involved.
2. Since we have to find generally the average speed (not the average time) so speeds are written on the top and corresponding time taken is written below.

Exp. 12) A mixture of rice is sold at ₹ 3.00 per kg. This mixture is formed by mixing the rice of ₹ 2.10 and ₹ 2.52 per kg. What is the ratio of price of cheaper to the costlier quality in the mixture if the profit of 25% is being earned.

(a) 5 : 2 (b) 2 : 7
(c) 2 : 5 (d) 15 : 8

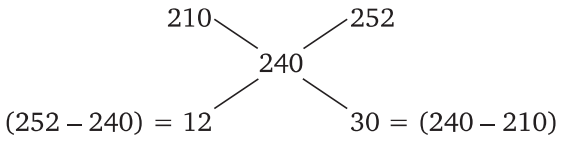
Solution Let the cost price of the mixture be ₹ *x* per kg, then

selling price $=x+\frac{25\times x}{100}=3$

$\Rightarrow 1.25x=3$

$\Rightarrow x=2.4$

Now the average cost price of mixture = ₹ 2.4



Therefore, the ratio of cheaper to costlier rice is 12 : 30, *i.e.*, 2 : 5.

NOTE ₹ 1 = 100 paise. There is no change in the ratio when we change the rupees into paise, just for our convenience in calculation.

Exp. 13) A milkman has 20 litres of milk. If he mixes 5 litres of water, which is freely available, in 20 litres of pure milk. If the cost of pure milk is ₹ 18 per litre, then the profit of the milkman, when he sells all the mixture at cost price, is :

(a) 20% (b) 25% (c) 33.33% (d) 18%

Solution When the water is freely available and all the water is sold at the price of the milk, then the water gives the profit on the cost of 20 litres of milk.

Therefore, profit percentage $= \frac{5}{20} \times 100 = 25\%$

Since, the profit % $= \frac{\text{profit}}{\text{cost price}} \times 100$

NOTE Here the milkman cheats his customer by false practise of creating illusion that instead of 20 litres milk, there is 25 litres of milk. So this extra 5 litres of milk (actually water) is the part of profit.

Exp. 14) In what ratio should freely available water and a premium priced wine be mixed so that after selling the mixture at the cost price a profit of 33.33% is made?

(a) 1 : 4 (b) 1 : 3 (c) 2 : 3 (d) 3 : 4

Solution 33.33% profit means there is one part water and 3 part is pure wine. So the required ratio of water and wine in the mixture is 1 : 3.

NOTE The above problem (No. 14) and the previous problem are basically related to the topics of ratio and profit-loss.

Exp. 15) In what ratio should freely available water be mixed with the wine worth ₹ 60 per litre so that after selling the mixture at ₹ 50 per litre, the profit will be 25%?

(a) 1 : 2 (b) 2 : 3 (c) 3 : 4 (d) 4 : 5

Solution Selling price = ₹ 50

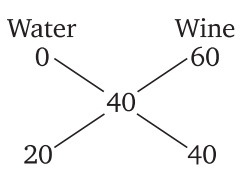
Therefore, the cost price

= ₹ 40, which is the average price.

It means the wine, worth ₹ 60 becomes worth ₹ 40 when the water was mixed in it. So we can conclude that in the mixture of ₹ 60, there is wine worth ₹ 40 and the rest is water.

Therefore, the ratio of water and wine is 20 : 40 *i.e.*, 1 : 2

Alternatively

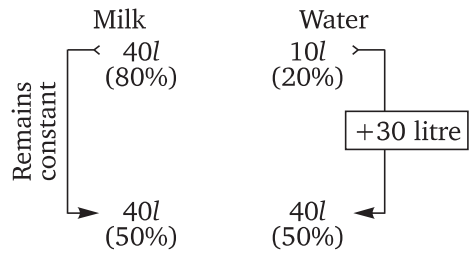


⇒ 1 : 2

Exp. 16) A mixture of water and milk contains 80% milk. In 50 litres of such a mixture, how many litres of water is required to increase the percentage of water to, 50%?

(a) 20
(b) 15
(c) 30
(d) none of the above

Solution Total amount of mixture is 50 litre



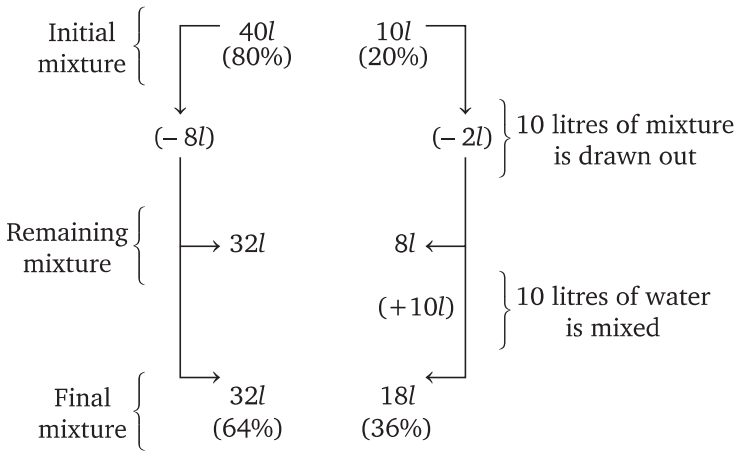
NOTE That, in this process the quantity of milk remains constant but its percentage decreases as the quantity (and percentage) of water increases.

Initially we have 40 litre milk and 10 litre water. Now we are required to have 50% water and remaining 50% milk in the new mixture. So we have to have 40 litre water, equal to the amount of pure milk (which is constant) available in the mixture. Thus we have to add up 30 litre (= 40 – 10) water in the original mixture.

Exp. 17) In a 50 litre mixture of water and milk, water is only 20%. The milkman gives 10 litres `of this' mixture to a customer and then he adds up 10 litres of pure water in the remaining mixture. The percentage of water in the final mixture is :

(a) 84% (b) 74%
(c) 26% (d) 36%

Solution Total quantity of mixture = 50 litre



See the chart for the Solution.

Initially there is 40 litres (80%) milk in the 50 litres mixture. When 10 litres of mixture is drawn out, it means 8 litres of milk is drawn out.

Now when the 10 litres of water is added, it means in the 50 litres of mixture, there is only 32 litres of milk, *i.e.*, 64%. Hence the percentage of water is 36%.

Exp. 18) There are three types of milk, Parag, Amul and Nestle. The ratio of fat to the non-fat contents in milk is 4 : 5, 5 : 6, 6 : 7 respectively. If all these three types of milk is mixed in equal quantity, the ratio of fat to the non-fat contents in the mixture will be :

(a) 1751 : 2110 (b) 175 : 543
(c) 3 : 5 (d) 10 : 18

Solution In this type of questions, we consider only one item (*viz.*, either fat or non-fat) as the fraction of the total quantity. Again we equate the denominators.
Let us consider fraction of 'fat'.

	Parag	Amul	Nestle
Fraction of fat →	$\frac{4}{9}$	$\frac{5}{11}$	$\frac{6}{13}$
	$\frac{4\times11\times13}{9\times11\times13}$	$\frac{5\times9\times13}{11\times9\times13}$	$\frac{6\times9\times11}{13\times9\times11}$
	$\frac{572}{1287}$	$\frac{585}{1287}$	$\frac{594}{1287}$

So the total fat in the total mixture

$$=\frac{572+585+594}{1287+1287+1287}=\frac{1751}{3861}$$

Hence, the ratio of fat to the non-fat contents in the mixture

$$=1751:2110 \qquad [\because\ 3861-1751=2110]$$

Exp. 19) Sharabi Chand purchased two different kinds of alcohol. In the first mixture the ratio of alcohol to water is 3 : 4 and in the second mixture it is 5 : 6. If he mixes the two given mixtures and makes a third mixture of 18 litres in which the ratio of alcohol to water is 4 : 5, the quantity of first mixture (whose ratio is 3 : 4) is required to make the 18 litres of the third kind of mixture is :

(a) 6 (b) 7 (c) 8 (d) 9

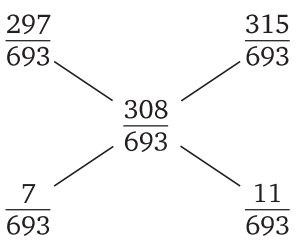
Solution The fraction of alcohol in the different mixtures is as follows :

$$\text{First mixture } \frac{3}{7}=\frac{297}{693}$$

$$\text{Second mixture } \frac{5}{11}=\frac{315}{693}$$

$$\text{Final mixture } \frac{4}{9}=\frac{308}{693}$$

Therefore,



Hence the ratio of first mixture is to second mixture is 7 : 11. Thus he has to mix 7 litres of first type of alcohol to make 18 litres of required mixture.

Exp. 20) Some amount out of ₹ 6000 was lent out at 10% per annum and the rest amount @ at 20% per annum and thus in 4 years the total interest from both the amounts collected was ₹ 3400. What is the amount which was lent out @ 10% per annum?

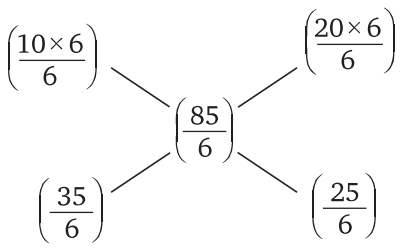
(a) ₹ 2500 (b) ₹ 2800
(c) ₹ 3200 (d) ₹ 3500

Solution The total interest of one year

$$=₹\ 850\ =\left(\frac{3400}{4}\right)$$

Therefore, the average rate of interest $=14\frac{1}{6}\%=\frac{85}{6}\%$

Thus,



Hence the ratio of amount which is lent @ 10% per annum to the amount lent @ 20% per annum is 35 : 25, *i.e.*, 7 : 5.Therefore, the amount which is lent out @ 10% per annum is ₹ 3500.

Exp. 21) From the 50 litres of milk, 5 litres of milk is taken out and after it 5 litres of water is added to the rest amount of milk. Again 5 litres of mixture of milk and water is drawn out and it was replaced by 5 litres of water. If this process is continued similarly for the third time, the amount of milk left after the third replacement :

(a) 45 *L* (b) 36.45 *L* (c) 40.5 *L* (d) 42.5 *L*

Solution

Milk	Water	
50 L	0 L	50 litre (initially)
− 5 L	− 0 L	withdrawn amount
45 L	0 L	50 litre (after first replacement)
+ 0 L	+ 5 L	
− 4.5 L	− 0.5 L	withdrawn amount
40.5 L	4.5 L	50 litre (after second replacement)
+ 0 L	+ 5 L	
− 4.05 L	− 0.95 L	withdrawn amount
36.45 L	8.55 L	50 litre (after third replacement)
	+ 5 L	

Shortcut

$$50\times\left(1-\frac{5}{50}\right)\times\left(1-\frac{5}{50}\right)\times\left(1-\frac{5}{50}\right)$$

$$=50\times\frac{45}{50}\times\frac{45}{50}\times\frac{45}{50}$$

$$=50\times\left(\frac{45}{50}\right)^3$$

$$=50\times\left(\frac{9}{10}\right)^3=36.45\ \text{L}$$

General Formula

Final or reduced concentration

= Initial concentration

$$\left(1-\frac{\text{amount being replaced in each operation}}{\text{total amount}}\right)^{n}$$

where n is the number of times the same operation is being repeated. The ''amount being replaced'' could be pure or mixture as per the case. Similarly, ''total amount'' could also be either pure or mixture. Here amount being replaced denotes the quantity which is to be withdrawn in each time.

Exp. 22) From a tank of petrol, which contains 200 litres of petrol, the seller replaces each time with kerosene when he sells 40 litres of petrol (or its mixture). Everytime he sells out only 40 litres of petrol (pure or impure). After replacing the petrol with kerosene 4th time, the total amount of kerosene in the mixture is :

(a) 81.92 *l* (b) 96 *l*

(c) 118.08 *l* (d) none of these

Solution The amount of petrol left after 4 operations

$$= 200\times\left(1-\frac{40}{200}\right)^{4} = 200\times\left(\frac{4}{5}\right)^{4}$$

$$= 200\times\frac{256}{625} = 81.92 \text{ litres}$$

Hence the amount of kerosene = 200 − 81.92 = 118.08 litres

Exp. 23) From a container of wine, a thief has stolen 15 litres of wine and replaced it with same quantity of water. He again repeated the same process. Thus in three attempts the ratio of wine and water became 343 : 169. The initial amount of wine in the container was :

(a) 75 litres (b) 100 litres

(c) 150 litres (d) 120 litres

Solution $\dfrac{\text{wine (left)}}{\text{water (added)}}=\dfrac{343}{169}$

It means $\dfrac{\text{wine (left)}}{\text{wine (initial amount)}}=\dfrac{343}{512}$ $(\because\ 343+169=512)$

Thus, $343x = 512x\left(1-\dfrac{15}{K}\right)^{3}$

$\Rightarrow$ $\dfrac{343}{512}=\left(\dfrac{7}{8}\right)^{3}=\left(1-\dfrac{15}{K}\right)^{3}$

$\Rightarrow$ $\left(1-\dfrac{15}{K}\right)=\dfrac{7}{8}=\left(1-\dfrac{1}{8}\right)$

$\Rightarrow$ $K = 120$

Thus the initial amount of wine was 120 litres.

Exp. 24) A jar was full with honey. A person used to draw out 20% of the honey from the jar and replaced it with sugar solution. He has repeated the same process 4 times and thus there was only 512 gm of honey left in the jar, the rest part of the jar was filled with the sugar solution. The initial amount of honey in the jar was :

(a) 1.25 kg (b) 1 kg

(c) 1.5 kg (d) none of these

Solution Let the initial amount of honey in the jar was K, then

$$512 = K\left(1-\frac{1}{5}\right)^{4} \qquad \left(\because\ 20\% = \frac{20}{100}=\frac{1}{5}\right)$$

or $512 = K\left(\dfrac{4}{5}\right)^{4}$

$\Rightarrow$ $K=\dfrac{512\times 625}{256}$

$\therefore$ $K = 1250$

Hence, initially the honey in the jar = 1.25 kg.

CAT-Test

Questions Helping you bell the CAT

1 How much Pepsi at ₹ 6 a litre is added to 15 litre of 'dew' at ₹ 10 a litre so that the price of the mixture be ₹ 9 a litre?
(a) 5 (b) 8
(c) 10 (d) none of these

2 In a municipal parking there are some two wheelers and rest are 4 wheelers. If wheels are counted, there are total 520 wheels but the incharge of the parking told me that there are only 175 vehicles. If no vehicle has a stepney, then the no. of two wheelers is :
(a) 75 (b) 100 (c) 90 (d) 85

3 In my pocket there are ₹ 25 consisting of only the denominations of 20 paise and 50 paise. Thus there are total 80 coins in my pocket. The no. of coins of the denomination of 50 paise is :
(a) 30 (b) 70 (c) 50 (d) 25

4 There are some shepherds and their sheep in a grazing field. The no. of total heads are 60 and total legs are 168 including both men and sheep. The no. of sheep is :
(a) 18 (b) 26 (c) 24 (d) 36

5 In the 75 litres of mixture of milk and water, the ratio of milk and water is 4 : 1. The quantity of water required to make the ratio of milk and water 3 : 1 is :
(a) 1 litre (b) 3 litres (c) 4 litres (d) 5 litres

6 In my office the average age of all the female employees is 21 years and that of male employees is 32 years, where the average age of all the (male and female) employees is 28 years. The total no. of employees in my office could be :
(a) 35 (b) 78 (c) 231 (d) 90

7 A car agency has 108 cars. He sold some cars at 9% profit and rest at 36% profit. Thus he gains 17% on the sale of all his cars. The no. of cars sold at 36% profit is :
(a) 25 (b) 32 (c) 35 (d) 75

8 ₹ 69 were divided among 115 students so that each girl gets 50 paise less than a boy. Thus each boy received twice the paise as each girl received. The no. of girls in the class is :
(a) 92 (b) 42 (c) 33 (d) 23

9 In what proportion water be mixed with spirit to gain 12.5% by selling it at cost price?
(a) 3 : 5 (b) 1 : 8 (c) 2 : 7 (d) 1 : 9

10 A butler stole wine from a butt of sherry containing 50% of spirit, then he replenished it by different wine containing 20% spirit. Thus there was only 30% strength (spirit) in the new mixture. How much of the original wine did he steal?
(a) 1/3 (b) 2/3 (c) 1/2 (d) 1/4

11 Mr. Mittal purchased two steel factories, one in India and other one in Malaysia for total ₹ 72 crores. Later on he sold the Indian factory at 16% profit and Malasian factory at 24% profit. Thus he gained a total profit of 19%. The selling price of Indian factory is :
(a) 45 crore (b) 52.2 crore
(c) 8.55 crore (d) can not be determined

12 In a 25 litre mixture of milk and water, the water is only 20%. How many litres of water is required to increase the percentage of water to 90%?
(a) 45 litre (b) 70 litre
(c) 115 litre (d) 175 litre

13 A milkman sells the milk at the cost price but he mixes the water (freely available) in it and thus he gains 9.09%. The quantity of water in the mixture of 1 litre is :
(a) 83.33 mL (b) 90.90 mL
(c) 99.09 mL (d) can't be determined

14 The price of petrol is ₹ 60 per litre and the price of spirit is ₹ 40 per litre. In what ratio the petrol and spirit be mixed such that the profit after selling the mixture at ₹ 75 per litre be 25%?
(a) 1 : 1
(b) 3 : 2
(c) 5 : 1
(d) such a mixture is not possible

15 A trader sells total 315 TV sets. He sells black and white TV sets at a loss of 6% and colour TV sets at a profit of 15%. Thus he gains 9% on the whole. The no. of B/W TV sets, which he has sold, is :
(a) 126 (b) 216 (c) 135 (d) 90

16 In a class of 30 students, the average weight of boys is 20 kg and the average weight of the girls is 25 kg. The fraction of boys out of the total students of the class is :
(a) $\frac{4}{5}$ (b) $\frac{5}{6}$
(c) $\frac{3}{4}$ (d) data insufficient

17 Baniya sells two types of tea *viz.* Desi Chai and Videshi Chai. He sells Desi Chai at ₹ 18 per kg and incurs a loss of 10% whereas on selling the Videshi Chai at ₹ 30 per kg, he gains 20%. In what proportion should the Desi Chai and Videshi Chai be mixed such that he can gain a profit of 25% by selling the mixture at ₹ 27.5 per kg?
(a) 3 : 2 (b) 2 : 3 (c) 2 : 5 (d) 3 : 5

18 The average age of boys in a class is 16.66, while the average age of girls is 18.75. Thus the average age of all the 40 students of the class is 17.5. If the difference between the no. of boys and girls is 8, then the no. of girls in the class is :
(a) 12 (b) 16
(c) 18 (d) Data insufficient

19 The ratio of water and alcohol in two different containers is 2 : 3 and 4 : 5. In what ratio we are required to mix the mixtures of two containers in order to get the new mixture in which the ratio of alcohol and water be 7 : 5 ?
(a) 7 : 3 (b) 5 : 3 (c) 8 : 5 (d) 2 : 7

20 The average marks of the students in four sections *A, B, C* and *D* together is 60%. The average marks of the students of *A*, *B*, *C* and *D* individually are 45%, 50%, 72% and 80% respectively. If the average marks of the students of sections *A* and *B* together is 48% and that of the students of *B* and *C* together is 60%. What is the ratio of number of students in sections *A* and *D*?
(a) 2 : 3 (b) 4 : 3 (c) 5 : 3 (d) 3 : 5

21 The diluted wine contains only 8 litres of wine and the rest is water. A new mixture whose concentration is 30%, is to be formed by replacing wine. How many litres of mixture shall be replaced with pure wine if there was initially 32 litres of water in the mixture ?
(a) 4 (b) 5
(c) 8 (d) none of these

22 The average weight of boys in a class is 30 kg and the average weight of girls in the same class is 20 kg. If the average weight of the whole class is 23.25 kg, what could be the possible strength of boys and girls respectively in the same class ?
(a) 14 and 26 (b) 13 and 27
(c) 17 and 27 (d) none of these

23 The shopkeeper mixed 40 kg refined oil with vegetable oil worth ₹ 60 per kg. Thus he gains ₹ 10 after selling the mixture of the two oils. The price of the first oil is :
(a) 20 (b) 25
(c) 45 (d) can't be determined

24. In a mixture of milk and water, there is only 26% water. After replacing the mixture with 7 litres of pure milk, the percentage of milk in the mixture become 76%. The quantity of mixture is :
(a) 65 litre (b) 91 litre
(c) 38 litre (d) none of these

25 The ratio of expenditure and savings is 3 : 2. If the income increases by 15% and the savings increases by 6%, then by how much per cent should his expenditure increases?
(a) 25 (b) 21
(c) 12 (d) 24

26 4 kg of a metal contains $\frac{1}{5}$ copper and rest is zinc. Another 5 kg of metal contains $\frac{1}{6}$ copper and rest is zinc. The ratio of copper and zinc into the mixture of these two metals :
(a) 49 : 221 (b) 39 : 231
(c) 94 : 181 (d) none of these

27 450 litres of a mixture of milk and water contain the milk and water in the ratio 9 : 1. How much water should be added to get a new mixture containing milk and water in the ratio 3 : 1?
(a) 54 (b) 90
(c) 45 (d) 63

28 The ratio of petrol and kerosene in the container is 3 : 2 when 10 litres of the mixture is taken out and is replaced by the kerosene, the ratio becomes 2 : 3. The total quantity of the mixture in the container is :
(a) 25 (b) 30
(c) 45 (d) cannot be determined

29 From a container, 6 litres milk was drawn out and was replaced by water. Again 6 litres of mixture was drawn out and was replaced by the water. Thus the quantity of milk and water in the container after these two operations is 9 : 16. The quantity of mixture is :
(a) 15 (b) 16
(c) 25 (d) 31

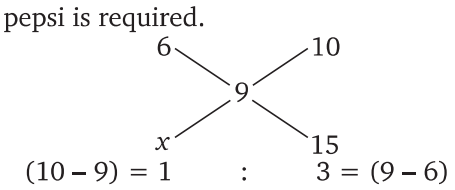
30 A milkman brings 100 litres pure milk from a dairy farmer and he sells 10 litres of it to the first customer, then he refills his vessel by adding 10 litres water. After this, he proceeds to the next house and sells 10 litres of it to the second customer and then he refills his vessel again by adding 10 litres of water. Thus, every time he sells 10 litres of milk - pure or impure - he keeps on replacing it with 10 litres of pure water.
Maximum how many customers can get at least 50% milk in the mixture that they purchase from this milkman?
(a) 5 (b) 6
(c) 7 (d) None of these

Answers

1 *(a)*	2. *(c)*	3. *(a)*	4. *(c)*	5. *(d)*	6. *(c)*	7. *(b)*	8. *(a)*	9. *(b)*	10. *(b)*
11. *(b)*	12. *(d)*	13. *(a)*	14. *(d)*	15. *(d)*	16. *(d)*	17. *(a)*	18. *(b)*	19. *(b)*	20. *(b)*
21. *(b)*	22. *(b)*	23. *(d)*	24. *(b)*	25. *(b)*	26. *(a)*	27. *(b)*	28. *(b)*	29. *(a)*	30. *(c)*

Hints & Solutions

1 Let x litre pepsi is required.



$(10 - 9) = 1 \quad : \quad 3 = (9 - 6)$

Therefore $\frac{x}{15} = \frac{1}{3} \Rightarrow x = 5$ litre

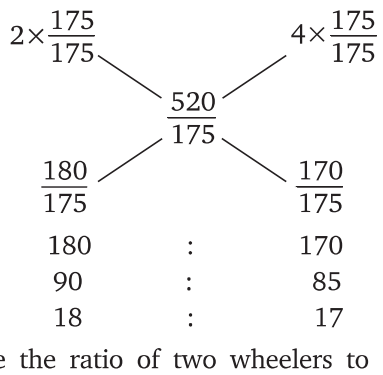
Alternatively Go through options.

2 Go through options : $90 \times 2 + 85 \times 4 = 520$

If 2 wheelers be 90 then the four wheelers will be

$85 = (175 - 90)$

Alternatively The average number of wheels per vehicle $= \frac{520}{175}$



$\Rightarrow \quad 180 \quad : \quad 170$

$\Rightarrow \quad 90 \quad : \quad 85$

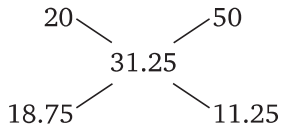
$\Rightarrow \quad 18 \quad : \quad 17$

Therefore the ratio of two wheelers to four wheelers is 18 : 17. Hence there are 90 "two wheelers".

3 Go through options : $30 \times 50 + 50 \times 20 = 2500$ paise

Alternatively Since the average price of a coin

$= \frac{2500}{80} = 31.25$ paise

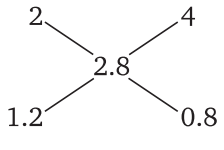


So the ratio of no. of 20 paise coins to the no. of 50 paise coins $= 18.75 : 11.25 = 75 : 45 = 5 : 3$

Therefore, the no. of coins of the denominations of 50 paise is 30.

4 Go through option : $24 \times 4 + 36 \times 2 = 168$

Alternatively



$\Rightarrow \quad 3 \quad : \quad 2$

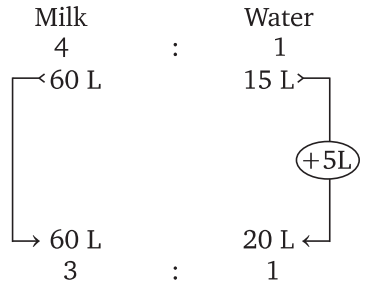
Therefore, the ratio of men and sheep is 3 : 2.

Alternatively Suppose there are only men, then the no. of legs $= 60 \times 2 = 120$.

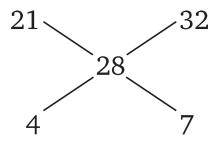
Now since there are $48 = (168 - 120)$ legs extra, it means there are $24 = \left(\frac{48}{2}\right)$ sheep, since a sheep has 2 extra legs than a man has.

5 Total quantity of mixture = 75 litre

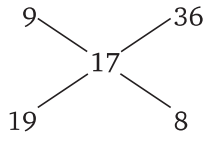
Therefore



6 Since the ratio of no. of female and male employees is 4 : 7, so the total no. of employees must be the multiples of 11. Hence the possible answer is 231.

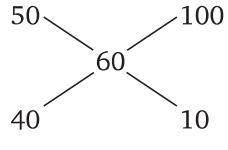


7 Since the ratio of cars sold at profit of 9% to the 36% is 19 : 8, hence the no. of cars sold at 36% profit is 32.



8 Here each girl receives 50 paise and each boy receives 100 paise and the average receiving of each student

$= \frac{6900}{115} = 60$ paise



$\Rightarrow \quad$ (G) 4 : (B) 1

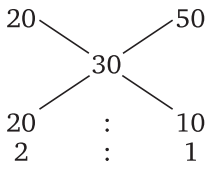
Thus the no. of girls = 92. [Number of boys = 23]

9 Profit $= 12.5\% = \frac{1}{8}$.

Hence the ratio of water to spirit is 1 : 8

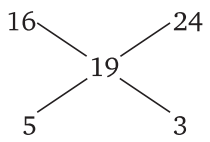
Since profit % $= \frac{\text{profit}}{\text{cost}} \times 100$.

10



Since the ratio of 20% wine to 50% wine is 2 : 1, it means there is $\frac{2}{3}$ wine which is replaced with wine in which the concentration of spirit is 20%.

11



Thus the cost price of Indian factory is ₹ 45 crore.
Therefore, the selling price of Indian factory is

$$= 45 + \frac{45 \times 16}{100} = 52.2 \text{ crore}$$

12

Milk		Water	
80%		20%	
4	:	1	
20 L		5 L	+ 175 L
20 L		180 L	
10%		90%	
1	:	9	

13 Profit (%) = 9.09% = $\frac{1}{11}$

Since the ratio of water and milk is 1 : 11,
Therefore the ratio of water is to mixture = 1 : 12
Thus the quantity of water in mixture of 1 litre

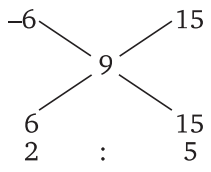
$$= 1000 \times \frac{1}{12} = 83.33 \text{ mL}$$

14 The selling price of mixture = ₹ 75

∴ The cost price of mixture = ₹ 60

Now we know that if he mixes the spirit (worth ₹ 40) with petrol (worth ₹ 60), the cost price of mixture must be less than ₹ 60, which is impossible. Hence there is no spirit with the petrol.

15



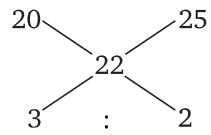
Thus the ratio of B/W TV sets to the no. of colour TV sets

$$= 2 : 5$$

Therefore, no. of B/W TV sets = 90

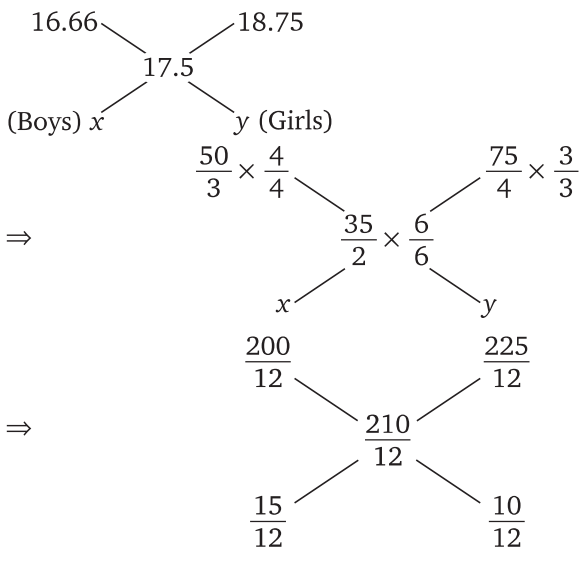
16 Since we do not know either the average weight of the whole class or the ratio of no. of boys to girls.

17 The S.P. of Desi Chai = ₹ 18
The S.P. of Videshi Chai = ₹ 30
The C.P. of Desi Chai = ₹ 20
The C.P. of Videshi Chai = ₹ 25
The S.P. of mixture = ₹ 27.5
The C.P. of mixture = ₹ 22



Therefore, the ratio of Desi Chai is to Videshi Chai is 3 : 2.

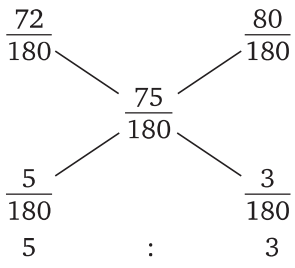
18



Thus the no. of girls = 16 and no. of boys = 24

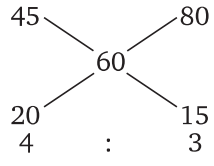
19

$W_1 : A_1$	$W_2 : A_2$	$W_N : A_N$
2 : 3	4 : 5	5 : 7
W_1 _ 2	W_2 _ 4	W_N _ 5
$W_1 + A_1$ – 5	$W_2 + A_2$ – 9	$W_N + A_N$ – 12
$= \frac{72}{180}$	$= \frac{80}{180}$	$= \frac{75}{180}$



Therefore, the ratio is 5 : 3.

20 Since the average marks of sections *B* and *C* together are equal the average marks of all the four sections (*i.e., A, B, C* and *D*), therefore the average marks of the remaining two sections *A* and *D* together will also be equal *i.e.,* 60%.



Hence, the required ratio is 4 : 3.

21

Wine		Water	
8 L		32 L	
1	:	4	
20%		80%	(original ratio)
30%		70%	(required ratio)

In this case, the percentage of water being reduced when the mixture is being replaced with wine.

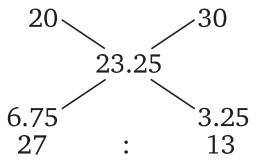
So the ratio of **left quantity** to the initial quantity is 7 : 8.

Therefore $\frac{7}{8}=\left[1-\frac{K}{40}\right] \Rightarrow \frac{7}{8}=\left[\frac{40-K}{40}\right]$

$\Rightarrow \quad K = 5$ litre

Alternatively Go through options.

22 Therefore no. of boys : Number of girls = 13 : 27



23 Since there is insufficient data.

24

Milk	Water	
74%	26%	(initially)
76%	24%	(after replacement)

$$\text{Left amount} = \text{Initial amount}\left(1-\frac{\text{replaced amount}}{\text{total amount}}\right)$$

$$24 = 26\left(1-\frac{7}{K}\right)$$

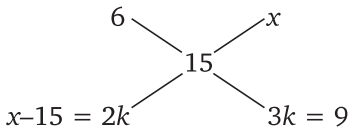
$$\Rightarrow \quad \frac{12}{13}=\left(1-\frac{7}{K}\right)$$

$$\Rightarrow \quad \frac{1}{13}=\frac{7}{K}$$

$$\Rightarrow \quad K = 91 \text{ litre}$$

NOTE In case of replacement, the initial (pure) amount is equal to the amount of final mixture.

25



Therefore $x = 21\%$.

26 Copper in $4 \text{ kg} = \frac{4}{5}$ kg

and Zinc in $4 \text{ kg} = 4 \times \frac{4}{5} = \frac{16}{5}$ kg

Copper in $5 \text{ kg} = 5 \times \frac{1}{6} = \frac{5}{6}$ kg

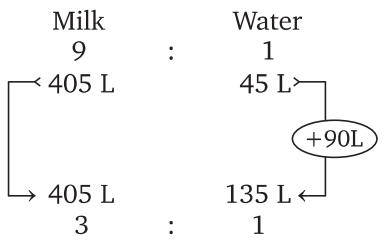
Zinc in $5 \text{ kg} = 5 \times \frac{5}{6} = \frac{25}{6}$ kg

Therefore, Copper in mixture $= \frac{4}{5}+\frac{5}{6}=\frac{49}{30}$ kg

and Zinc in the mixture $= \frac{16}{5}+\frac{25}{6}=\frac{221}{30}$ kg

Therefore, the required ratio = 49 : 221.

27



28. Petrol : Kerosene

3 : 2 (initially)

2 : 3 (after replacement)

$$\frac{\text{Remaining (or left) quantity}}{\text{Initial quantity}}=\left(1-\frac{\text{replaced quantity}}{\text{total quantity}}\right)$$

$$\text{(for petrol)}\ \frac{2}{3}=\left(1-\frac{10}{K}\right) \Rightarrow \frac{1}{3}=\frac{10}{K}$$

$\Rightarrow \quad K = 30$ litre

Therefore, the total quantity of the mixture in the container is 30 litre.

29. $\frac{9}{25}=\left(1-\frac{6}{K}\right)^2 \Rightarrow \frac{3}{5}=\left(1-\frac{6}{K}\right)$

$\Rightarrow \quad K = 15$ litre

30. After the first customer, who gets absolutely pure milk, every next customer gets less pure milk than the previous customer. The purity of milk for the nth customer who receives only impure milk can be calculated as following.

$$50 \geq 100\left(1-\frac{10}{100}\right)^n$$

The highest value of n that satisfies the above relation is 6. Therefore, we have 6 customers who get milk, which is less than 100% pure but more than 50% pure. However, there is one customer who gets 100% pure milk. So, there are actually 7 customers who get more than 50% pure milk.

Hence, choice (c) is the correct one.

Alternatively We can do it manually, if we are really good at calculation.

The First customer gets 100% pure milk.

The Second customer gets 90% pure milk.

The Third customer gets 81% pure milk.

The Fourth customer gets 72.9% pure milk.

The Fifth customer gets 65.61% pure milk.

The Sixth customer gets 59.05% pure milk.

The Seventh customer gets 53.14% pure milk.

The eighth customer gets 47.83% pure milk.

Thus, we see that maximum 7 customers get at least 50% pure milk.